## Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) start from the observation that connectedness is central to risk measurement (market, credit and systemic risk) but is poorly defined, and that correlation-based measures capture only pairwise, linear association. They propose to measure connectedness with the variance decomposition of an approximating model: $d_{ij}^H$ is the share of the $H$-step forecast-error variance of variable $i$ due to shocks in variable $j$. The matrix $D^H = [d_{ij}^H]$, augmented with its row and column sums, forms a *connectedness table*. Its off-diagonal entries are pairwise directional connectedness from $j$ to $i$; their row sums measure total directional connectedness *from* others to $i$, their column sums connectedness *to* others from $j$, and the difference gives net connectedness, analogous to a trade balance. Total connectedness is the average off-diagonal row (or column) sum, a single system-wide number. Because shocks are correlated, they use the generalized variance decomposition of Koop et al. (1996) and Pesaran and Shin (1998), normalized so that each row sums to one, which avoids the ordering dependence of Cholesky identification.

The paper's second contribution is conceptual: a variance-decomposition matrix is a weighted, directed network. "From" and "to" connectedness are the node from-degrees and to-degrees, and total connectedness is the mean degree of the network. The authors also relate the two directions to systemic-risk measures: from-degrees resemble exposure measures such as marginal expected shortfall, while to-degrees resemble contribution measures such as $\Delta$CoVaR.

Empirically, they study 13 large US financial institutions from May 1999 to April 2010, using daily realized volatilities computed from five-minute intraday returns (in logs), a VAR(3) approximating model, a 12-day horizon and rolling windows of 100 days. Full-sample total connectedness is 78.3%; connectedness "from" others is tightly distributed (70–82%), whereas connectedness "to" others ranges from 53% to 106%, with Citigroup the largest net transmitter. The rolling index shows two large cycles, one from 2000 to 2003 (the end of the dot-com boom, September 11 and the WorldCom bankruptcy) and one during the 2007–2009 crisis, when it rose from about 56% in February 2007 to close to 90% in August 2007 and reached its maximum of 89.2% after the Lehman bankruptcy. Results are robust to windows of 75 and 125 days, horizons of 6 and 18 days and Cholesky orderings. Our application changes the reference universe (emerging-market country funds instead of US financial firms), the object (returns instead of volatilities) and the frequency (weekly instead of daily).

NETS (Barigozzi and Brownlees, 2019) estimates the network directly as a sparse object: a lasso-type estimator of a VAR delivers a *Granger* network (lagged dependence), the sparse concentration matrix of the VAR innovations a *contemporaneous* partial-correlation network, and their combination a *long-run* partial-correlation network. FNETS (Barigozzi et al., 2024) first removes a common factor structure, estimated by dynamic principal components in a generalized dynamic factor model (Forni et al., 2000), and then estimates the same sparse networks for the idiosyncratic component, separating connectedness due to pervasive common shocks from idiosyncratic interdependence.

# 1 Motivation

Emerging equity markets are exposed to two kinds of shocks. Some are global: changes in US monetary policy, in the dollar or in investors' appetite for risk move all of them at once, a pattern that Rey (2013) describes as a global financial cycle. Others are local, such as an election, a currency crisis or a change in economic policy in one country, and the question is whether, and through which countries, they spread to the rest. Telling the two apart matters for several reasons. For investors, the benefit of diversifying across emerging markets depends on how much of their co-movement is a common component that no portfolio of these markets can diversify away. For policy makers, a high degree of connectedness calls for different responses if

it reflects bilateral transmission (contagion) or a common exposure to the same external shock. For the empirical literature, the distinction is an old one: Forbes and Rigobon (2002) showed that much apparent contagion during crises is high interdependence driven by common shocks and higher volatility.

The tools to measure connectedness have multiplied, and they do not all measure the same thing. The variance-decomposition approach of Diebold and Yılmaz (2014) summarizes how much of each market's forecast-error variance is due to shocks elsewhere; it is dense by construction and does not separate common from bilateral transmission. Sparse network estimators such as NETS (Barigozzi and Brownlees, 2019) keep only the links supported by the data, and FNETS (Barigozzi et al., 2024) first removes the common factor and then estimates the network of what is left. Applied to the same data, these methods can give quite different pictures of how connected markets are, who transmits and who receives, and how stable the network is. Before using a connectedness measure to draw conclusions about contagion, it is therefore worth knowing how much of the measured connectedness depends on the estimation method.

Our sample of 17 emerging markets offers clear examples of both kinds of shock. In the week of the Argentine primary elections of August 2019 the Argentine fund (ARGT) lost 28.4% (in log terms), its largest weekly fall in the sample: a large, country-specific shock. Seven months later, the COVID-19 crash of March 2020 produced the largest weekly losses of almost every market at the same time: 14 of the 17 markets recorded their worst week in March 2020. Argentina is one case among the 17 markets in this paper, but its 2019 episode illustrates the question we ask: whether a large local shock spreads to other markets, and whether the methods agree on that.

## 2 Introduction

This paper asks how measured connectedness among emerging equity markets changes with the method used to estimate it, and what share of it reflects a global common factor rather than idiosyncratic links between markets. We study weekly returns of US-listed country funds for 17 emerging markets in Latin America, Asia and EMEA, together with the S&P 500 as a global reference, from March 2011 to October 2026, and we estimate connectedness in four ways: the variance-decomposition measures of Diebold and Yılmaz (2014) on a VAR estimated by OLS; the same measures on a sparse VAR estimated by the lasso with time-series cross-validation; NETS (Barigozzi and Brownlees, 2019), which estimates sparse Granger, contemporaneous and long-run partial-correlation networks; and FNETS (Barigozzi et al., 2024), which estimates the same networks after removing the common factor. All methods are estimated on the same rolling windows and mapped into comparable measures, and we compare them on three criteria: the density of the networks, the agreement between methods, and their stability over time.

We test three hypotheses. First, regularization alone should matter little for total connectedness at the weekly frequency, because the cross-market dependence of returns is mostly contemporaneous rather than lagged. Second, a substantial share of the connectedness measured by Diebold–Yilmaz should be due to a common factor, so that it falls markedly once the factor is removed, and its time variation should follow the weight of that factor. Third, sparse estimators should produce networks that are less dense, and also less stable from one window to the next, than the dense variance-decomposition network.

The results support the first two hypotheses and only partly the third. The lasso VAR sets almost all lag coefficients to zero yet gives nearly the same connectedness as Diebold–Yilmaz (80.0% against 81.0% on average over windows, correlation 0.98). NETS gives a lower level (61.5%) and keeps about half of the possible long-run links, while tracking the Diebold–Yilmaz series closely (correlation 0.89). FNETS finds a single common factor in every window, which explains on average 58% of the variance of the returns; removing it lowers connectedness by 41% on average, a share that is stable across window lengths, horizons, the exclusion of the S&P 500

and the exclusion of 2020. Over time, Diebold–Yilmaz connectedness and the weight of the common factor move together. The sparse networks are indeed much less dense, but stability depends on the measure: NETS connectedness jumps more from one window to the next than Diebold–Yilmaz, yet its strongest long-run links are more persistent; the idiosyncratic network of FNETS is both the sparsest and the least stable. South Africa, Korea and Taiwan are the main net transmitters and Turkey, Vietnam and Argentina the main net receivers in the dense and NETS systems; once the common factor is removed, Argentina is close to neutral.

Our contribution is to put the base paper's connectedness measures side by side with sparse and factor-adjusted network estimators on the same data, windows and metrics, for a set of emerging markets, and to quantify the part of measured connectedness that is a common factor. Section 3 describes the data, Section 4 the methods and comparison criteria, Section 5 the results, and Sections 6 and 7 discuss them and conclude. Robustness checks are in the appendix.

# 3 Data

**Universe and sample.** We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks, with no missing values.

**Returns and frequency.** Prices are dividend- and split-adjusted, so returns are total returns. We use weekly Friday-to-Friday log returns. Choosing the weekly frequency is a decision of ours, for three reasons: weekly returns retain some cross-market predictability, which is the dynamic dependence a VAR-based network is meant to capture; weekly aggregation mitigates the non-synchronous trading between Asian and EMEA local markets and their US-listed ETFs; and it reduces the microstructure noise of daily ETF prices. We also choose to study returns rather than volatilities. Diebold and Yılmaz (2014) measure the connectedness of realized volatilities; with daily prices the same could be done here, by building weekly realized volatilities, but we leave volatility connectedness as an extension and focus on the transmission of return shocks. Returns are not devoid of cross-market predictive content: in a full-sample VAR(1) on the 18 series, 59 of the 324 slope coefficients (18.2%) are significant at the 5% level, against 5% expected by chance. Our networks therefore describe the transmission of return shocks, not of risk.

**Descriptive statistics.** Table 1 reports summary statistics. All series are strongly leptokurtic and mostly negatively skewed; an augmented Dickey–Fuller test rejects a unit root for every series. Five weekly log returns exceed 25% in absolute value (ARGT in the week of the August 2019 primary elections and four observations in March 2020); they are genuine market moves and are kept. The March 2020 crash dominates the tails: of the ten largest weekly drops pooled across the 17 emerging markets, eight occur in the weeks ending 13 and 20 March 2020, and 14 of the 17 markets record their worst week in March 2020.

**Limitations.** (i) ETF returns embed exchange-rate movements against the USD and are subject to non-synchronous trading. (ii) FXI tracks the FTSE China 50 Index, i.e. 50 large-cap Chinese companies listed in Hong Kong, and EPI tracks the earnings-weighted WisdomTree